

Sequential Task:

The Magic Backpack

Object-Oriented Programming and GUI

March 23, 2026

1. Copy all `.java` files attached to the assignment and place them in the packages of a newly created project. Preserve consistency between the declared package names and the directory structure in the project.
2. The class `Main.java` does not compile because the variable `backpack` cannot be used in a `for` loop.
 - (a) Declare the class `BackpackIterator`, which implements the interface `Iterator<Item>` and allows iteration over backpack items while skipping worthless ones.
 - i. Add a private, immutable field `items` of type `Item[]`, which stores a reference to the backpack item array.
 - ii. Add a private, immutable field `itemCount`, which stores the number of actually occupied elements in the array.
 - iii. Add a private field `currentIndex`, which stores the current iteration position.
 - iv. Implement the constructor `BackpackIterator(Item[] items, int itemCount)`, whose purpose is to initialize the iterator and set the starting position.
 - Assign the passed array `items` to the field `items`.
 - Assign the passed value `itemCount` to the field `itemCount`.
 - Set the field `currentIndex` to 0.
 - Call the helper method `skipWorthless()` to skip any worthless elements at the beginning.
 - v. Implement the private method `skipWorthless()`, whose purpose is to move the iteration index to the nearest valuable item.
 - Run a `while` loop as long as `currentIndex` is less than `itemCount` and the element at the current index satisfies the condition `isWorthless()`.
 - In each iteration, increment `currentIndex` to skip the worthless item.
 - vi. Implement the method `hasNext()`, whose purpose is to check whether there is another valuable item to return.
 - Return the boolean value resulting from the comparison `currentIndex < itemCount`.
 - vii. Implement the method `next()`, whose purpose is to return the next valuable item.

- Call the method `hasNext()` and check whether another element is available.
 - If there are no more elements, throw a `NoSuchElementException` with a message stating that there are no more valuable items.
 - Retrieve the element from the array `items` at index `currentIndex` and store it in the variable `item`.
 - Increment `currentIndex` by one.
 - Call the method `skipWorthless()` to prepare the iterator for the next call.
 - Return the retrieved element `item`.
- (b) Go to the class `MagicBackpack` and make sure that it implements the `Iterable` interface for the generic type `Item` in section *TODO 01*.
- (c) Go to section *TODO 02* and implement the method `iterator()`, whose purpose is to provide an iterator that allows traversal of the backpack items.
- i. Create a new instance of the class `BackpackIterator`, passing the current array `items` and the number of occupied elements `itemCount` to its constructor.
 - ii. Return the created object as the method result, so that iteration over the elements is possible using the interface `Iterator<Item>`.
3. The final result should look as follows:

THE ADVENTURER'S MAGIC BACKPACK

Packing the backpack...

Added to backpack: Health Potion
 Added to backpack: Broken Bottle
 Added to backpack: Silver Sword
 Added to backpack: Old Boot
 Added to backpack: Ruby
 Added to backpack: Mana Potion
 Added to backpack: Wet Rag
 Added to backpack: Crossbow
 Added to backpack: Ancient Coin
 Added to backpack: Poison Vial
 Added to backpack: Diamond

=====

Backpack of: Geralt of Rivia

Weight: 10.0 / 30.0 kg | Items: 11 (valuable: 8, junk: 3)

Total value: 2325 gold

Hero sees (magic filter):

1. Health Potion | value: 50 gold | weight: 0.3 kg [Restores health]

2. Silver Sword	value: 500 gold weight: 3.5 kg [dmg:45, dur:3]
3. Ruby	value: 300 gold weight: 0.1 kg [Rare]
4. Mana Potion	value: 45 gold weight: 0.3 kg [Restores mana]
5. Crossbow	value: 200 gold weight: 4.0 kg [dmg:30, dur:5]
6. Ancient Coin	value: 150 gold weight: 0.1 kg [Epic]
7. Poison Vial	value: 80 gold weight: 0.2 kg [Deals damage]
8. Diamond	value: 1000 gold weight: 0.1 kg [Legendary]

The backpack hides 3 junk items...

=====

Full backpack contents (debug):

1. Health Potion	value: 50 gold weight: 0.3 kg [Restores health]
2. Broken Bottle	value: 0 gold weight: 0.2 kg [Shards of glass]
3. Silver Sword	value: 500 gold weight: 3.5 kg [dmg:45, dur:3]
4. Old Boot	value: 0 gold weight: 0.8 kg [Smells terrible]
5. Ruby	value: 300 gold weight: 0.1 kg [Rare]
6. Mana Potion	value: 45 gold weight: 0.3 kg [Restores mana]
7. Wet Rag	value: 0 gold weight: 0.5 kg [Completely worthless]
8. Crossbow	value: 200 gold weight: 4.0 kg [dmg:30, dur:5]
9. Ancient Coin	value: 150 gold weight: 0.1 kg [Epic]
10. Poison Vial	value: 80 gold weight: 0.2 kg [Deals damage]
11. Diamond	value: 1000 gold weight: 0.1 kg [Legendary]

Using items...

You drink Health Potion! Restores health (+75 pts)

Health Potion - the bottle is already empty!

You attack with Silver Sword! Damage: 45 (durability: 2)

You attack with Silver Sword! Damage: 45 (durability: 1)

You attack with Silver Sword! Damage: 45 The weapon broke!

Silver Sword - the weapon is destroyed!

You drink Poison Vial! Deals damage (+100 pts)

Searching for 'Ruby'...

Found: Ruby	value: 300 gold weight: 0.1 kg [Rare]
-------------	---

Items that can still be used:

->Mana Potion

->Crossbow

```
Hero browses the backpack (enhanced for):  
(junk is automatically hidden)
```

```
I see: Health Potion (value: 50 gold)  
I see: Silver Sword (value: 500 gold)  
I see: Ruby (value: 300 gold)  
I see: Mana Potion (value: 45 gold)  
I see: Crossbow (value: 200 gold)  
I see: Ancient Coin (value: 150 gold)  
I see: Poison Vial (value: 80 gold)  
I see: Diamond (value: 1000 gold)
```

End of adventure!

4.  Milestone 1: present the result of your work to the instructor
5. Next, let us address the issue of sorting the items in the backpack.
 - (a) In the class `Main.java`, find section *TODO 03*. Uncomment the code below it, and replace the three dots with a call to the no-argument method `sort` on the variable `backpack`.
 - (b) Go to the class `Item`, section *TODO 04*, and make sure that it implements the interface `Comparable` with the generic parameter `Item`.
 - (c) In section *TODO 05*, implement the method `compareTo(Item other)`, whose purpose is to define the natural ordering of objects of type `Item` based on their value and name.
 - i. Compare the values of the two `Item` objects using the method `Integer.compare`.
 - Pass the value of the object `other.value` as the first argument.
 - Pass the value of the current object `this.value` as the second argument.
 - This will produce descending order by value.
 - ii. Store the comparison result in the helper variable `result`.
 - iii. Check whether the result of the value comparison is equal to 0, which means that both objects have the same value.
 - If the values are equal, perform an additional comparison of the names.
 - Call the method `compareToIgnoreCase` on the field `name`, passing the name of the object `other.name` as the argument.
 - Assign the result of this comparison to the variable `result`.
 - iv. Return the value of the variable `result` as the method result, defining the ordering relation between the objects.
6. The final result should look as follows:

THE ADVENTURER'S MAGIC BACKPACK

Packing the backpack...

Added to backpack: Health Potion
Added to backpack: Broken Bottle
Added to backpack: Silver Sword
Added to backpack: Old Boot
Added to backpack: Ruby
Added to backpack: Mana Potion
Added to backpack: Wet Rag
Added to backpack: Crossbow
Added to backpack: Ancient Coin
Added to backpack: Poison Vial
Added to backpack: Diamond

=====

Backpack of: Geralt of Rivia

Weight: 10.0 / 30.0 kg | Items: 11 (valuable: 8, junk: 3)

Total value: 2325 gold

Hero sees (magic filter):

1. Health Potion	value: 50 gold weight: 0.3 kg [Restores health]
2. Silver Sword	value: 500 gold weight: 3.5 kg [dmg:45, dur:3]
3. Ruby	value: 300 gold weight: 0.1 kg [Rare]
4. Mana Potion	value: 45 gold weight: 0.3 kg [Restores mana]
5. Crossbow	value: 200 gold weight: 4.0 kg [dmg:30, dur:5]
6. Ancient Coin	value: 150 gold weight: 0.1 kg [Epic]
7. Poison Vial	value: 80 gold weight: 0.2 kg [Deals damage]
8. Diamond	value: 1000 gold weight: 0.1 kg [Legendary]

The backpack hides 3 junk items...

=====

Full backpack contents (debug):

1. Health Potion	value: 50 gold weight: 0.3 kg [Restores health]
2. Broken Bottle	value: 0 gold weight: 0.2 kg [Shards of glass]
3. Silver Sword	value: 500 gold weight: 3.5 kg [dmg:45, dur:3]
4. Old Boot	value: 0 gold weight: 0.8 kg [Smells terrible]
5. Ruby	value: 300 gold weight: 0.1 kg [Rare]
6. Mana Potion	value: 45 gold weight: 0.3 kg [Restores mana]

7. Wet Rag	value: 0 gold weight: 0.5 kg [Completely worthless]
8. Crossbow	value: 200 gold weight: 4.0 kg [dmg:30, dur:5]
9. Ancient Coin	value: 150 gold weight: 0.1 kg [Epic]
10. Poison Vial	value: 80 gold weight: 0.2 kg [Deals damage]
11. Diamond	value: 1000 gold weight: 0.1 kg [Legendary]

Natural sort (most expensive first)...

Backpack of: Geralt of Rivia

Weight: 10.0 / 30.0 kg | Items: 11 (valuable: 8, junk: 3)

Total value: 2325 gold

Hero sees (magic filter):

1. Diamond	value: 1000 gold weight: 0.1 kg [Legendary]
2. Silver Sword	value: 500 gold weight: 3.5 kg [dmg:45, dur:3]
3. Ruby	value: 300 gold weight: 0.1 kg [Rare]
4. Crossbow	value: 200 gold weight: 4.0 kg [dmg:30, dur:5]
5. Ancient Coin	value: 150 gold weight: 0.1 kg [Epic]
6. Poison Vial	value: 80 gold weight: 0.2 kg [Deals damage]
7. Health Potion	value: 50 gold weight: 0.3 kg [Restores health]
8. Mana Potion	value: 45 gold weight: 0.3 kg [Restores mana]

The backpack hides 3 junk items...

=====

Using items...

You drink Health Potion! Restores health (+75 pts)

Health Potion - the bottle is already empty!

You attack with Silver Sword! Damage: 45 (durability: 2)

You attack with Silver Sword! Damage: 45 (durability: 1)

You attack with Silver Sword! Damage: 45 The weapon broke!

Silver Sword - the weapon is destroyed!

You drink Poison Vial! Deals damage (+100 pts)

Searching for 'Ruby'...

Found: Ruby | value: 300 gold | weight: 0.1 kg [Rare]

Items that can still be used:

```
->Crossbow
->Mana Potion
```

Hero browses the backpack (enhanced for):
(junk is automatically hidden)

```
I see: Diamond (value: 1000 gold)
I see: Silver Sword (value: 500 gold)
I see: Ruby (value: 300 gold)
I see: Crossbow (value: 200 gold)
I see: Ancient Coin (value: 150 gold)
I see: Poison Vial (value: 80 gold)
I see: Health Potion (value: 50 gold)
I see: Mana Potion (value: 45 gold)
```

End of adventure!

7.  Milestone 2: **present the result of your work to the instructor**
8. Declare the class `ItemNameComparator`, which implements the interface `Comparator<Item>` and defines how objects of type `Item` are compared based on their names.
 - (a) Override the method `compare(Item a, Item b)`, whose purpose is to determine the order of two `Item` objects.
 - Retrieve the names of both objects by calling the method `getName()`.
 - Compare the names using the method `compareToIgnoreCase` to obtain alphabetical order without considering letter case.
 - Return the comparison result as a value of type `int`, defining the ordering relation between the objects.
9. In the class `Main.java`, find section *TODO 06*. Uncomment the code below it, and replace the three dots with a call to the method `sort` on the variable `backpack`, passing an object of type `ItemNameComparator` as the argument.
10. The final result should look as follows:

THE ADVENTURER'S MAGIC BACKPACK

Packing the backpack...

```
Added to backpack: Health Potion
Added to backpack: Broken Bottle
Added to backpack: Silver Sword
```

Added to backpack: Old Boot
Added to backpack: Ruby
Added to backpack: Mana Potion
Added to backpack: Wet Rag
Added to backpack: Crossbow
Added to backpack: Ancient Coin
Added to backpack: Poison Vial
Added to backpack: Diamond

=====
Backpack of: Geralt of Rivia
Weight: 10.0 / 30.0 kg | Items: 11 (valuable: 8, junk: 3)
Total value: 2325 gold
=====

Hero sees (magic filter):

1. Health Potion	value: 50 gold weight: 0.3 kg [Restores health]
2. Silver Sword	value: 500 gold weight: 3.5 kg [dmg:45, dur:3]
3. Ruby	value: 300 gold weight: 0.1 kg [Rare]
4. Mana Potion	value: 45 gold weight: 0.3 kg [Restores mana]
5. Crossbow	value: 200 gold weight: 4.0 kg [dmg:30, dur:5]
6. Ancient Coin	value: 150 gold weight: 0.1 kg [Epic]
7. Poison Vial	value: 80 gold weight: 0.2 kg [Deals damage]
8. Diamond	value: 1000 gold weight: 0.1 kg [Legendary]

The backpack hides 3 junk items...

=====
Full backpack contents (debug):

1. Health Potion	value: 50 gold weight: 0.3 kg [Restores health]
2. Broken Bottle	value: 0 gold weight: 0.2 kg [Shards of glass]
3. Silver Sword	value: 500 gold weight: 3.5 kg [dmg:45, dur:3]
4. Old Boot	value: 0 gold weight: 0.8 kg [Smells terrible]
5. Ruby	value: 300 gold weight: 0.1 kg [Rare]
6. Mana Potion	value: 45 gold weight: 0.3 kg [Restores mana]
7. Wet Rag	value: 0 gold weight: 0.5 kg [Completely worthless]
8. Crossbow	value: 200 gold weight: 4.0 kg [dmg:30, dur:5]
9. Ancient Coin	value: 150 gold weight: 0.1 kg [Epic]
10. Poison Vial	value: 80 gold weight: 0.2 kg [Deals damage]
11. Diamond	value: 1000 gold weight: 0.1 kg [Legendary]

=====

Natural sort (most expensive first)...

Backpack of: Geralt of Rivia

Weight: 10.0 / 30.0 kg | Items: 11 (valuable: 8, junk: 3)

Total value: 2325 gold

=====

Hero sees (magic filter):

1. Diamond	value: 1000 gold weight: 0.1 kg [Legendary]
2. Silver Sword	value: 500 gold weight: 3.5 kg [dmg:45, dur:3]
3. Ruby	value: 300 gold weight: 0.1 kg [Rare]
4. Crossbow	value: 200 gold weight: 4.0 kg [dmg:30, dur:5]
5. Ancient Coin	value: 150 gold weight: 0.1 kg [Epic]
6. Poison Vial	value: 80 gold weight: 0.2 kg [Deals damage]
7. Health Potion	value: 50 gold weight: 0.3 kg [Restores health]
8. Mana Potion	value: 45 gold weight: 0.3 kg [Restores mana]

The backpack hides 3 junk items...

=====

Alphabetical sort (Comparator)...

Backpack of: Geralt of Rivia

Weight: 10.0 / 30.0 kg | Items: 11 (valuable: 8, junk: 3)

Total value: 2325 gold

=====

Hero sees (magic filter):

1. Ancient Coin	value: 150 gold weight: 0.1 kg [Epic]
2. Crossbow	value: 200 gold weight: 4.0 kg [dmg:30, dur:5]
3. Diamond	value: 1000 gold weight: 0.1 kg [Legendary]
4. Health Potion	value: 50 gold weight: 0.3 kg [Restores health]
5. Mana Potion	value: 45 gold weight: 0.3 kg [Restores mana]
6. Poison Vial	value: 80 gold weight: 0.2 kg [Deals damage]
7. Ruby	value: 300 gold weight: 0.1 kg [Rare]
8. Silver Sword	value: 500 gold weight: 3.5 kg [dmg:45, dur:3]

The backpack hides 3 junk items...

=====

Using items...

You drink Health Potion! Restores health (+75 pts)

Health Potion - the bottle is already empty!

You attack with Silver Sword! Damage: 45 (durability: 2)
You attack with Silver Sword! Damage: 45 (durability: 1)
You attack with Silver Sword! Damage: 45 The weapon broke!
Silver Sword - the weapon is destroyed!

You drink Poison Vial! Deals damage (+100 pts)

Searching for 'Ruby'...

Found: Ruby | value: 300 gold | weight: 0.1 kg [Rare]

Items that can still be used:

->Crossbow
->Mana Potion

Hero browses the backpack (enhanced for):
(junk is automatically hidden)

I see: Ancient Coin (value: 150 gold)
I see: Crossbow (value: 200 gold)
I see: Diamond (value: 1000 gold)
I see: Health Potion (value: 50 gold)
I see: Mana Potion (value: 45 gold)
I see: Poison Vial (value: 80 gold)
I see: Ruby (value: 300 gold)
I see: Silver Sword (value: 500 gold)

End of adventure!

11. Declare the class `ItemWeightComparator`, which implements the interface `Comparator<Item>` and defines how objects of type `Item` are compared based on their weight and value.
 - (a) Override the method `compare(Item a, Item b)`, whose purpose is to determine the order of two `Item` objects.
 - i. Compare the weights of both objects using the method `Double.compare`.
 - Pass the value `a.getWeight()` as the first argument.
 - Pass the value `b.getWeight()` as the second argument.
 - This will produce ascending order by weight.
 - ii. Store the comparison result in the variable `result`.
 - iii. Check whether the result of the weight comparison is equal to 0, which means that both objects have the same weight.
 - If the weights are equal, perform an additional comparison of their values.

- Call the method `Integer.compare`, passing `b.getValue()` as the first argument and `a.getValue()` as the second.
 - In this way, obtain descending order by value for items with the same weight.
 - Assign the result of this comparison to the variable `result`.
- iv. Return the value of the variable `result` as the result of the method.
12. In the class `Main.java`, find section *TODO 07*. Uncomment the code below it, and replace the three dots with a call to the method `sort` on the variable `backpack`, passing an object of type `ItemWeightComparator` as the argument.
13. The final result should look as follows:

THE ADVENTURER'S MAGIC BACKPACK

Packing the backpack...

Added to backpack: Health Potion
 Added to backpack: Broken Bottle
 Added to backpack: Silver Sword
 Added to backpack: Old Boot
 Added to backpack: Ruby
 Added to backpack: Mana Potion
 Added to backpack: Wet Rag
 Added to backpack: Crossbow
 Added to backpack: Ancient Coin
 Added to backpack: Poison Vial
 Added to backpack: Diamond

Backpack of: Geralt of Rivia

Weight: 10.0 / 30.0 kg | Items: 11 (valuable: 8, junk: 3)

Total value: 2325 gold

Hero sees (magic filter):

1. Health Potion	value: 50 gold weight: 0.3 kg [Restores health]
2. Silver Sword	value: 500 gold weight: 3.5 kg [dmg:45, dur:3]
3. Ruby	value: 300 gold weight: 0.1 kg [Rare]
4. Mana Potion	value: 45 gold weight: 0.3 kg [Restores mana]
5. Crossbow	value: 200 gold weight: 4.0 kg [dmg:30, dur:5]
6. Ancient Coin	value: 150 gold weight: 0.1 kg [Epic]
7. Poison Vial	value: 80 gold weight: 0.2 kg [Deals damage]
8. Diamond	value: 1000 gold weight: 0.1 kg [Legendary]

```

The backpack hides 3 junk items...
=====

Full backpack contents (debug):
  1. Health Potion      | value: 50 gold | weight: 0.3 kg [Restores health]
  2. Broken Bottle      | value: 0 gold | weight: 0.2 kg [Shards of glass]
  3. Silver Sword       | value: 500 gold | weight: 3.5 kg [dmg:45, dur:3]
  4. Old Boot           | value: 0 gold | weight: 0.8 kg [Smells terrible]
  5. Ruby               | value: 300 gold | weight: 0.1 kg [Rare]
  6. Mana Potion        | value: 45 gold | weight: 0.3 kg [Restores mana]
  7. Wet Rag            | value: 0 gold | weight: 0.5 kg [Completely worthless]
  8. Crossbow           | value: 200 gold | weight: 4.0 kg [dmg:30, dur:5]
  9. Ancient Coin       | value: 150 gold | weight: 0.1 kg [Epic]
 10. Poison Vial        | value: 80 gold | weight: 0.2 kg [Deals damage]
 11. Diamond            | value: 1000 gold | weight: 0.1 kg [Legendary]

Natural sort (most expensive first)...

Backpack of: Geralt of Rivia
Weight: 10.0 / 30.0 kg | Items: 11 (valuable: 8, junk: 3)
Total value: 2325 gold

Hero sees (magic filter):
  1. Diamond            | value: 1000 gold | weight: 0.1 kg [Legendary]
  2. Silver Sword       | value: 500 gold | weight: 3.5 kg [dmg:45, dur:3]
  3. Ruby               | value: 300 gold | weight: 0.1 kg [Rare]
  4. Crossbow           | value: 200 gold | weight: 4.0 kg [dmg:30, dur:5]
  5. Ancient Coin       | value: 150 gold | weight: 0.1 kg [Epic]
  6. Poison Vial        | value: 80 gold | weight: 0.2 kg [Deals damage]
  7. Health Potion      | value: 50 gold | weight: 0.3 kg [Restores health]
  8. Mana Potion        | value: 45 gold | weight: 0.3 kg [Restores mana]

The backpack hides 3 junk items...
=====

Alphabetical sort (Comparator)...

Backpack of: Geralt of Rivia
Weight: 10.0 / 30.0 kg | Items: 11 (valuable: 8, junk: 3)
Total value: 2325 gold

```

Hero sees (magic filter):

1. Ancient Coin	value: 150 gold weight: 0.1 kg [Epic]
2. Crossbow	value: 200 gold weight: 4.0 kg [dmg:30, dur:5]
3. Diamond	value: 1000 gold weight: 0.1 kg [Legendary]
4. Health Potion	value: 50 gold weight: 0.3 kg [Restores health]
5. Mana Potion	value: 45 gold weight: 0.3 kg [Restores mana]
6. Poison Vial	value: 80 gold weight: 0.2 kg [Deals damage]
7. Ruby	value: 300 gold weight: 0.1 kg [Rare]
8. Silver Sword	value: 500 gold weight: 3.5 kg [dmg:45, dur:3]

The backpack hides 3 junk items...

=====

Weight sort (Comparator)...

Backpack of: Geralt of Rivia

Weight: 10.0 / 30.0 kg | Items: 11 (valuable: 8, junk: 3)

Total value: 2325 gold

Hero sees (magic filter):

1. Diamond	value: 1000 gold weight: 0.1 kg [Legendary]
2. Ancient Coin	value: 150 gold weight: 0.1 kg [Epic]
3. Ruby	value: 300 gold weight: 0.1 kg [Rare]
4. Poison Vial	value: 80 gold weight: 0.2 kg [Deals damage]
5. Health Potion	value: 50 gold weight: 0.3 kg [Restores health]
6. Mana Potion	value: 45 gold weight: 0.3 kg [Restores mana]
7. Silver Sword	value: 500 gold weight: 3.5 kg [dmg:45, dur:3]
8. Crossbow	value: 200 gold weight: 4.0 kg [dmg:30, dur:5]

The backpack hides 3 junk items...

=====

Using items...

You drink Health Potion! Restores health (+75 pts)

Health Potion - the bottle is already empty!

You attack with Silver Sword! Damage: 45 (durability: 2)

You attack with Silver Sword! Damage: 45 (durability: 1)

You attack with Silver Sword! Damage: 45 The weapon broke!

Silver Sword - the weapon is destroyed!

You drink Poison Vial! Deals damage (+100 pts)

Searching for 'Ruby'...

Found: Ruby | value: 300 gold | weight: 0.1 kg [Rare]

Items that can still be used:

->Mana Potion

->Crossbow

Hero browses the backpack (enhanced for):

(junk is automatically hidden)

I see: Diamond (value: 1000 gold)

I see: Ancient Coin (value: 150 gold)

I see: Ruby (value: 300 gold)

I see: Poison Vial (value: 80 gold)

I see: Health Potion (value: 50 gold)

I see: Mana Potion (value: 45 gold)

I see: Silver Sword (value: 500 gold)

I see: Crossbow (value: 200 gold)

End of adventure!

14. ★ Milestone 3 present the result of your work to the instructor